Mateo Albrecht

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Education

Texas A&M University College Station, TX
Master of Science in Microelectronics and Semiconductors – Incoming Graduate Student *Fall 2027*

Texas A&M University College Station, TX
Bachelor of Science in Computer Engineering, Minor in Mathematics *August 2023 – May 2027*

- GPA: 3.6/4.0
- Relevant Coursework: Digital Integrated Circuit Design, Microprocessor System Design, Computer Architecture, Computer Systems, Electronics

Experience

Part-Time Software Engineer May 2026 – Present
PowerDB

- Develop and maintain web application features using HTML, JavaScript, and Node.js for PowerDB's web platform
- Integrate frontend components with backend services to support application data flow and user-facing workflows
- Diagnose defects and refactor existing UI components to improve reliability, maintainability, and application behavior

Projects

Senior Capstone – MIL-STD-1750A Processor August 2026 – May 2027
Verilog, Yosys, OpenROAD, OpenSTA, SKY130 *ASIC / RTL Design*

- Developing a synthesizable 16-bit MIL-STD-1750A-compatible processor with a five-stage pipeline in a three-member capstone team, targeting SKY130 ASIC implementation
- Designing and verifying 16-bit instruction and data memory RTL; creating testbenches for instruction fetch, memory reads, and clocked writes and integrating with processor datapath and control interfaces
- Applying Yosys, OpenROAD, and OpenSTA for logic synthesis, static timing analysis (STA), area evaluation, and development of an RTL-to-GDSII implementation flow

IoT Smart Light Controller
ESP32, C++, Firebase, REST APIs, Swift, CAD

- Engineered an ESP32-based motorized lighting system with embedded C++ control logic and Firebase cloud synchronization for remote operation
- Developed iOS automation in Swift and designed a custom 3D-printed servo mount, validating end-to-end hardware-software integration

Organization & Publication

Texas A&M NASA RASC-AL Team – Machine Learning Team August 2024 – May 2026

- Developed and evaluated YOLOv8 computer vision, PID control, and Gazebo autonomy pipelines for autonomous lunar-rover navigation and payload operations
- Co-authored conference publication *Navigation and Optimized Support Rover*, published in 2025

Technical Skills

Languages & HDL : Verilog, SystemVerilog, C, C++, Python, JavaScript, HTML/CSS, Java, Swift, SQL
Digital Design & EDA : RTL Design, RTL Verification, Testbench Development, Logic Synthesis, Static Timing Analysis (STA), Yosys, OpenROAD, OpenSTA, Vivado, Cadence Virtuoso, SKY130
Embedded & Hardware : ESP32, Raspberry Pi, Arduino, Zybo Z7-10 FPGA, Circuit Design, Electronics, 3D Printing/CAD
Tools & Platforms : Git, Linux, Docker, VS Code, Visual Studio, Xcode, Firebase, REST APIs
Languages : English, Spanish